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TRAFFIC ENGINEERING

METI

Lab 2 Report [EN]

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Group 5

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1 Introduction

The objective of this laboratory assignment was to implement and analyze IP routing mechanisms using Cisco routers within the GNS3 simulation environment.

The experiments performed throughout the laboratory focused on dynamic routing protocols, routing behavior, path selection mechanisms, and routing convergence.

The laboratory activities included:

- IPv4 subnetting using VLSM;
- router interface configuration;
- OSPF routing configuration;
- routing table analysis;
- OSPF metric manipulation;
- connectivity verification;
- link failure simulation and convergence analysis.

The network topology consisted of five routers (R1 to R5) and two end devices (PC1 and PC2), interconnected through multiple redundant paths.

2 Part A — OSPF Network Configuration

2.1 Network Topology

The network topology implemented in GNS3 is presented below.

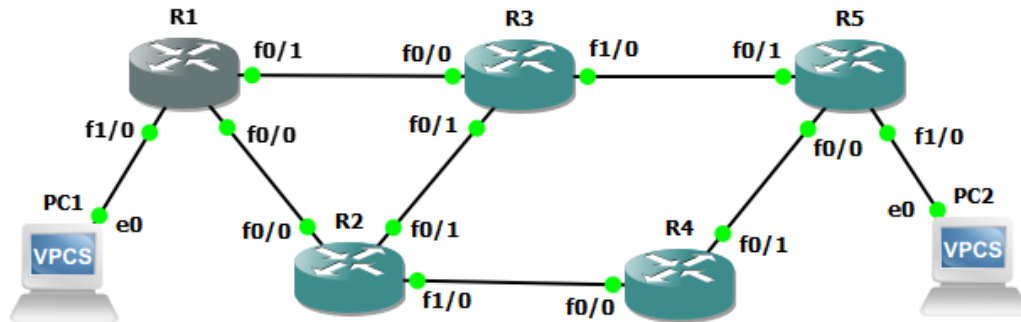


Figure 1: General network topology

The topology contains five routers interconnected through multiple redundant paths, allowing OSPF to dynamically calculate alternative routes in case of failures.

Additionally:

- PC1 is connected to R1;
- PC2 is connected to R5.

2.1.1 Broadcast Domains

Each physical link corresponds to an independent subnet/broadcast domain.

The network contains:

- 6 router-to-router subnets;
- 2 LAN subnets (PC1-R1 and PC2-R5).

Therefore, the topology contains a total of:

8 broadcast domains

2.2 IPv4 Addressing Plan

The network address block assigned was:

192.168.5.0/24

Using VLSM, the address space was segmented into multiple /30 subnets.

Link	Subnet	IPs Used
R1-R2	192.168.5.4/30	.5 / .6
R1-R3	192.168.5.8/30	.9 / .10
R2-R3	192.168.5.12/30	.13 / .14
R2-R4	192.168.5.16/30	.17 / .18
R4-R5	192.168.5.20/30	.21 / .22
R3-R5	192.168.5.24/30	.25 / .26
PC1-R1	192.168.5.0/30	.1 / .2
PC2-R5	192.168.5.28/30	.29 / .30

Table 1: IPv4 addressing plan

2.3 Router Configuration

Each router interface was configured manually.

Example configuration used on R1:

```
interface FastEthernet1/0
ip address 192.168.5.2 255.255.255.252
no shutdown
```

```
interface FastEthernet0/0
ip address 192.168.5.5 255.255.255.252
no shutdown
```

```
interface FastEthernet0/1
ip address 192.168.5.9 255.255.255.252
no shutdown
```

The command below was used to verify interface status:

```
show ip interface brief
```

All interfaces reached the up/up operational state.

The same configuration methodology was applied to all routers in the topology.

2.4 OSPF Configuration

OSPF was configured using process ID 1.

The following commands were applied:

```
router ospf 1
network 192.168.5.0 0.0.0.255 area 0
```

After configuration, OSPF neighbor relationships were successfully established.

The command used to verify adjacency status was:

```
show ip ospf neighbor
```

Routers reached the FULL adjacency state, confirming successful synchronization of OSPF databases.

All routers were configured within OSPF Area 0.

2.5 Routing Table Analysis

The routing table was inspected using:

```
show ip route
```

Routes learned dynamically through OSPF were identified with the letter:

O

Initially, multiple equal-cost paths existed for some destinations, demonstrating Equal Cost Multi-Path (ECMP) behavior.

Example:

```
O 192.168.5.12 [110/20] via 192.168.5.10
                via 192.168.5.6
```

This indicates that two paths with identical OSPF cost were available.

This demonstrated the capability of OSPF to support load balancing across equal-cost paths.

2.6 OSPF Cost Manipulation

To analyze OSPF path selection behavior, the cost of interface FastEthernet0/1 on R1 was increased.

Configuration used:

```
interface FastEthernet0/1
ip ospf cost 100
```

After modifying the cost:

- OSPF recalculated the routing table;
- alternative lower-cost paths were selected;
- traffic was redirected automatically.

Traceroute confirmed the path modification.

Before: Equal-cost paths existed through both R2 and R3.

$$R1 \rightarrow R2 \rightarrow R3 \rightarrow R5$$

After: Traffic preferred the path through R2.

$$R1 \rightarrow R3 \rightarrow R5$$

This demonstrated OSPF metric-based path selection.

This experiment demonstrated how OSPF metrics directly influence path selection decisions.

2.7 Experimental Results

This section presents the experimental results obtained during the implementation and testing of the network.

2.7.1 Interface Verification

After configuring the interfaces, the following command was used to verify interface status:

```
show ip interface brief
```

Example output from R1:

Interface	IP-Address	Status	Protocol
FastEthernet0/0	192.168.5.5	up	up
FastEthernet0/1	192.168.5.9	up	up
FastEthernet1/0	192.168.5.2	up	up

The output confirmed that all interfaces were correctly configured and operational.

2.7.2 OSPF Neighbor Adjacency

The following command was used to verify OSPF neighbor relationships:

```
show ip ospf neighbor
```

Observed output:

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.5.25	1	FULL/BDR	00:00:32	192.168.5.10	FastEthernet0/1
192.168.5.17	1	FULL/DR	00:00:33	192.168.5.6	FastEthernet0/0

The FULL state confirmed that OSPF adjacencies were successfully established and routing databases were synchronized.

2.7.3 Routing Table Analysis

The routing table was analyzed using:

```
show ip route
```

Example output:

```
O 192.168.5.12 [110/20] via 192.168.5.10
                    via 192.168.5.6

O 192.168.5.24 [110/11] via 192.168.5.10

O 192.168.5.16 [110/11] via 192.168.5.6
```

Routes marked with the letter **O** correspond to routes learned through OSPF.

Initially, some routes had multiple equal-cost paths, demonstrating Equal Cost Multi-Path (ECMP) behavior.

2.7.4 OSPF Cost Manipulation

To influence path selection, the OSPF cost of interface FastEthernet0/1 on R1 was increased. Configuration used:

```
interface FastEthernet0/1
ip ospf cost 100
```

After changing the cost, the routing table changed accordingly.

Updated routing example:

```
O 192.168.5.12 [110/20] via 192.168.5.6

O 192.168.5.24 [110/21] via 192.168.5.6

O 192.168.5.20 [110/21] via 192.168.5.6
```

This confirmed that OSPF selected the path with the lowest accumulated metric.

2.7.5 Connectivity Verification

Connectivity between end devices was tested using ICMP echo requests.

Command executed:

```
ping 192.168.5.30
```

Observed output:

```
!!!!
Success rate is 100 percent (5/5)
```

This confirmed successful communication between PC1 and PC2.

Traceroute was also performed:

```
traceroute 192.168.5.30
```


Observed route before OSPF cost manipulation:

```
1 192.168.5.6
2 192.168.5.14
3 192.168.5.26
4 192.168.5.30
```

After increasing the OSPF cost, the selected route changed:

```
1 192.168.5.10
2 192.168.5.26
3 192.168.5.30
```

This demonstrated successful dynamic route recalculation.

2.7.6 Link Failure and OSPF Convergence

To evaluate convergence behavior, one of the active interfaces was disabled.

Command executed:

```
interface FastEthernet0/0
shutdown
```

OSPF immediately detected the failure:

```
%OSPF-5-ADJCHG:
Process 1, Nbr 192.168.5.17 on FastEthernet0/0
from FULL to DOWN
```

The routing table was automatically recalculated and traffic was redirected through alternative paths.

Traceroute after failure:

```
1 192.168.5.10
2 192.168.5.26
3 192.168.5.30
```

The interface was restored using:

```
no shutdown
```

OSPF adjacency was re-established successfully:

```
%OSPF-5-ADJCHG:
Process 1, Nbr 192.168.5.17
from LOADING to FULL
```

This demonstrated:

- automatic fault detection;
- dynamic rerouting;
- successful OSPF convergence.

3 Part B — MPLS Configuration and Analysis

3.1 Introduction

The objective of Part B was to configure and analyze an MPLS-based network using Cisco routers in GNS3.

The experiments focused on:

- enabling MPLS forwarding;
- configuring LDP (Label Distribution Protocol);
- analyzing MPLS forwarding tables;
- observing MPLS labels using Wireshark;
- demonstrating Penultimate Hop Popping (PHP);
- testing MPLS recovery after link failures.

The MPLS network was implemented on top of the OSPF infrastructure configured in Part A.

3.2 MPLS Configuration

Cisco Express Forwarding (CEF) was enabled on all routers to support MPLS operation.

The following commands were applied:

```
ip cef
mpls ip
```

MPLS was enabled on the router interfaces participating in the MPLS core.

Example configuration on R3:

```
interface FastEthernet0/0
mpls ip
```

```
interface FastEthernet1/0
mpls ip
```

```
interface FastEthernet1/1
mpls ip
```

OSPF was used as the underlying routing protocol to distribute reachability information between routers.

3.3 LDP Neighbor Verification

LDP neighbors were verified using the following command:

```
show mpls ldp neighbor
```

Example output obtained on R3:

```
Peer LDP Ident: 192.168.5.9:0
State: Oper
```

```
Peer LDP Ident: 192.168.5.29:0
State: Oper
```

```
Peer LDP Ident: 192.168.5.17:0
State: Oper
```

The **Oper** state confirmed that LDP sessions were successfully established and labels were being exchanged between MPLS routers.

3.4 MPLS Interface Verification

MPLS-enabled interfaces were verified using:

```
show mpls interfaces
```

Example output:

Interface	IP	Operational
FastEthernet0/0	Yes (ldp)	Yes
FastEthernet1/0	Yes (ldp)	Yes
FastEthernet1/1	Yes (ldp)	Yes

This confirmed that MPLS forwarding and LDP were operational on all core links.

3.5 MPLS Forwarding Table Analysis

The MPLS forwarding table was inspected using:

```
show mpls forwarding-table
```

Example output obtained on R3:

Local Label	Outgoing Label	Prefix
16	Pop Label	192.168.5.4/30
17	Pop Label	192.168.5.0/30
18	Pop Label	192.168.5.16/30
19	Pop Label	192.168.5.20/30
20	Pop Label	192.168.5.28/30

The forwarding table demonstrated that MPLS labels were correctly assigned and distributed across the network.

The presence of the Pop Label operation confirmed the implementation of Penultimate Hop Popping (PHP).

3.6 Wireshark Analysis and PHP Demonstration

Wireshark captures were performed to analyze MPLS label switching behavior.

Traffic was generated using:

```
ping 192.168.5.30
```

and:

```
trace 192.168.5.30
```

3.6.1 MPLS Labels in the Core Network

Wireshark captures performed on core MPLS links showed packets containing MPLS headers.

The captured packets contained entries similar to:

```
MultiProtocol Label Switching Header
Label: 20
```

This confirmed that MPLS labels were being attached and switched correctly inside the MPLS core.

No.	Time	Source	Destination	Protocol	Length	Info
17	10.177612	192.168.5.1	192.168.5.30	ICMP	102	Echo (ping) request id=0xd8f8, seq=1/256, ttl=63 (reply in 18)
18	10.213598	192.168.5.30	192.168.5.1	ICMP	98	Echo (ping) reply id=0xd8f8, seq=1/256, ttl=62 (request in 17)
22	17.227854	192.168.5.1	192.168.5.30	ICMP	102	Echo (ping) request id=0xd9f8, seq=2/512, ttl=63 (reply in 23)
23	17.264063	192.168.5.30	192.168.5.1	ICMP	98	Echo (ping) reply id=0xd9f8, seq=2/512, ttl=62 (request in 22)
25	18.274848	192.168.5.1	192.168.5.30	ICMP	102	Echo (ping) request id=0xdaf8, seq=3/768, ttl=63 (reply in 26)
26	18.302451	192.168.5.30	192.168.5.1	ICMP	98	Echo (ping) reply id=0xdaf8, seq=3/768, ttl=62 (request in 25)
27	19.315375	192.168.5.1	192.168.5.30	ICMP	102	Echo (ping) request id=0xdbf8, seq=4/1024, ttl=63 (reply in 28)
28	19.350761	192.168.5.30	192.168.5.1	ICMP	98	Echo (ping) reply id=0xdbf8, seq=4/1024, ttl=62 (request in 27)
29	20.362878	192.168.5.1	192.168.5.30	ICMP	102	Echo (ping) request id=0xdcf8, seq=5/1280, ttl=63 (reply in 30)
30	20.389362	192.168.5.30	192.168.5.1	ICMP	98	Echo (ping) reply id=0xdcf8, seq=5/1280, ttl=62 (request in 29)

Frame 17: 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface -, id 0 Ethernet II, Src: ca:01:27:78:00:1d (ca:01:27:78:00:1d), Dst: ca:02:27:f6:00:00 (ca:02:27:f6:00:00) MultiProtocol Label Switching Header, Label: 20, Exp: 0, S: 1, TTL: 63 Internet Protocol Version 4, Src: 192.168.5.1, Dst: 192.168.5.30 Internet Control Message Protocol Type: 8 (Echo (ping) request) Code: 0 Checksum: 0x4712 [correct] [Checksum Status: Good] Identifier (BE): 55544 (0xd8f8) Identifier (LE): 63704 (0xf8d8) Sequence Number (BE): 1 (0x0001) Sequence Number (LE): 256 (0x0100) [Response frame: 18] Data (56 bytes)	
--	--

Figure 2: MPLS labeled packet captured in the MPLS core

3.6.2 Penultimate Hop Popping (PHP)

Additional captures were performed on the final hop between R5 and PC2.

The captured packets no longer contained MPLS headers and only displayed IP and ICMP information.

This demonstrated that the MPLS label was removed by the penultimate router before reaching the egress router.

Therefore, Penultimate Hop Popping (PHP) was successfully verified.

No.	Time	Source	Destination	Protocol	Length	Info
3	6.226199	192.168.5.1	192.168.5.30	ICMP	98	Echo (ping) request id=0xd8f8, seq=1/256, ttl=61 (reply in 6)
6	6.236952	192.168.5.30	192.168.5.1	ICMP	98	Echo (ping) reply id=0xd8f8, seq=1/256, ttl=64 (request in 3)
7	7.280991	192.168.5.1	192.168.5.30	ICMP	98	Echo (ping) request id=0xd9f8, seq=2/512, ttl=61 (reply in 8)
8	7.281746	192.168.5.30	192.168.5.1	ICMP	98	Echo (ping) reply id=0xd9f8, seq=2/512, ttl=64 (request in 7)
9	8.321180	192.168.5.1	192.168.5.30	ICMP	98	Echo (ping) request id=0xdaf8, seq=3/768, ttl=61 (reply in 10)
10	8.323252	192.168.5.30	192.168.5.1	ICMP	98	Echo (ping) reply id=0xdaf8, seq=3/768, ttl=64 (request in 9)
11	9.367700	192.168.5.1	192.168.5.30	ICMP	98	Echo (ping) request id=0xdbf8, seq=4/1024, ttl=61 (reply in 12)
12	9.367771	192.168.5.30	192.168.5.1	ICMP	98	Echo (ping) reply id=0xdbf8, seq=4/1024, ttl=64 (request in 11)
14	10.410128	192.168.5.1	192.168.5.30	ICMP	98	Echo (ping) request id=0xdcf8, seq=5/1280, ttl=61 (reply in 15)
15	10.410224	192.168.5.30	192.168.5.1	ICMP	98	Echo (ping) reply id=0xdcf8, seq=5/1280, ttl=64 (request in 14)


```

Frame 3: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
  Ethernet II, Src: ca:04:28:34:00:1c (ca:04:28:34:00:1c), Dst: Private_66:68:01 (00:50:79:66:68:01)
  Internet Protocol Version 4, Src: 192.168.5.1, Dst: 192.168.5.30
  Internet Control Message Protocol
    Type: 8 (Echo (ping) request)
    Code: 0
    Checksum: 0x4712 [correct]
    [Checksum Status: Good]
    Identifier (BE): 55544 (0xd8f8)
    Identifier (LE): 63704 (0xf8d8)
    Sequence Number (BE): 1 (0x0001)
    Sequence Number (LE): 256 (0x0100)
    [Response frame: 6]
  Data (56 bytes)

```

Figure 3: Packet capture after PHP operation (no MPLS label present)

3.7 Connectivity Verification

Connectivity between PC1 and PC2 was verified using ICMP echo requests.

The following command was executed:

```
ping 192.168.5.30
```

Observed output:

```
!!!!
Success rate is 100 percent (5/5)
```

This confirmed successful MPLS forwarding and end-to-end connectivity.

Traceroute was also performed:

```
trace 192.168.5.30
```

Observed route before link failure:

```
1 192.168.5.2
2 192.168.5.10
3 192.168.5.26
4 192.168.5.30
```

This path corresponded to:

$$R1 \rightarrow R3 \rightarrow R5$$

3.8 MPLS Failure Recovery and Convergence

To evaluate MPLS recovery behavior, the primary MPLS path between R1 and R3 was disabled.

The following command was applied on R1:

```
interface FastEthernet1/1
shutdown
```

After the failure:

- OSPF recalculated the routing table;
- LDP sessions were updated;
- MPLS forwarding switched to an alternative LSP.

Traceroute after the failure produced the following output:

```
1 192.168.5.2
2 192.168.5.6
3 192.168.5.14
4 192.168.5.26
5 192.168.5.30
```

The new path corresponded to:

$$R1 \rightarrow R2 \rightarrow R3 \rightarrow R5$$

This demonstrated successful MPLS rerouting and network resilience after link failure.

The failed interface was restored using:

```
interface FastEthernet1/1
no shutdown
```

After convergence, the original LSP was automatically restored.

3.9 Conclusion

This part of the laboratory successfully demonstrated the deployment and operation of MPLS forwarding in a Cisco-based network.

The experiments confirmed:

- successful LDP neighbor establishment;
- MPLS label distribution and forwarding;
- operation of Penultimate Hop Popping (PHP);
- MPLS traffic analysis using Wireshark;
- MPLS recovery after link failures.

The obtained results demonstrated the robustness and flexibility of MPLS networks under dynamic routing conditions.

Overall, the laboratory provided practical experience with MPLS forwarding, label switching, and traffic engineering concepts.

4 Conclusion

This laboratory assignment successfully demonstrated the deployment and analysis of an OSPF-based IP network using Cisco routers in GNS3.

The implementation of VLSM enabled efficient IPv4 address allocation across all network segments. OSPF dynamically exchanged routing information and automatically selected optimal paths according to interface metrics.

Several experiments were performed to evaluate OSPF behavior under different network conditions, including:

- routing table analysis;
- OSPF cost manipulation;
- connectivity verification;
- link failure simulation;
- routing convergence analysis.

The obtained results demonstrated:

- successful end-to-end connectivity;
- dynamic path recalculation;
- automatic failover capabilities;
- fast OSPF convergence.

Overall, this assignment provided practical experience with dynamic routing protocols and fundamental traffic engineering concepts.